# Bureaucratic Climate Change: An analysis of the relationship between United States Executive Office and Drought Severity

Amanda Giddings

Stat7100

Date: 4/30/2025

Abstract

With each presidency, there is a series of executive orders that will be signed and many of those orders will have an environmental impact of some form. Due to this, I wanted to analyze the environmental impact that is drought severity with elected presidents from both the Democratic and Republican Parties. This is not only an analysis on the presidents, but also an analysis on how the voting patterns of Americans affect the very environment that we live in. I run independence tests on the variables to determine whether there is a relationship between political parties and drought severity and then determine what that relationship looks like. I have found that there is a positive correlation between the two, with Republicans having an influx of drought severity and Democrats efflux of drought severity.

## Introduction

Droughts across the United States have become more frequent with the increase of climate change globally. However, is there something that can be done on our soil to combat this issue? Is this an issue that is unique to specific regions in the United States and do climate initiatives that are implemented by our executive administration play a role in the severity of the droughts that we experience?

This analysis was created to review the presidential candidates during their time in office to see whether their choice in policy positively or negatively affected our environment. I wanted to see if the policies that are implemented either led to increased droughts or decreased droughts in terms of severity and time they affected people.

This analysis will not only benefit policymakers when determining what policies should be passed, but it will also benefit voters. In order for voters to make educated decisions on who they are putting in executive office, they need to consider the effects that the policies implemented will have on the environment that they live in. If there is a candidate that is more likely to harm voters with their environmental policies, this is something that voters should know since they are the primary stakeholders. It also has an effect on the industries throughout the United States and can aid all parties in preparing for droughts depending on the party in office.

Data Selection and Preparation

**Dataset Overview:**

The data was downloaded from the drought.gov website and included the overall area percentage of all 6 climate regions throughout the continental United States. For that data, it is logged every week, so I drew data from the first week of a president's full start term and the last week of a president's full start term (i.e. for Obama I pulled weekly data for January to December 2009 and weekly data for January to December 2016.) These datasets included 53 weeks with 6 different drought severities for all 6 different climate regions. That means the initial dataset was roughly 15, 264 individual observations. The drought severities were categorized using a nominal system where none was the lowest and D4 was the highest.

I chose to evaluate the Biden, Trump, Obama, and Bush administrations to give both an accurate timeframe for both administration types as well as an accurate representation of each administration type since covid influenced both the Biden and Trump administrations.

**Categorical Variables:**

Administration – Either "Republican" or "Democrat"

Drought Severity – None = Normal or wet conditions, D0 = abnormally dry/entering or exiting a drought, D1 = Moderate, D2 = Severe, D3 = Extreme, and D4 = Exceptional

Climate Region – Midwest, Northeast, Plains, South, Southeast, West

Administrative Years – First Year and Last Year for each administrative term

Week – Measured 1 – 53 to for each week of the first and last year that a president is in office

**Quantitative Variables:**

Area Percent – Measured by the percentage of the total land area affected by droughts of each type

Weeks Affected - Number of weeks that a certain drought severity persisted

Drought Number – The total number of times/the frequency of a drought occurrence by severity type

Area Percentage Total – Measured by the percentage of the area of land affected by drought by first year, last year, republican, and democrat

Rep Tot (Republican Total) – F_Tot (first year total – singular) and L_Tot (last year total – singular)

Dem Tot (Democrats Total) – F_Tot (first year total – singular) and L_Tot (last year total – singular)

FY (First Year Total) – Week_Tot (sum of first year of Republican terms and Democrat terms)

LY (Last Year Total) – Week_Tot (sum of last year of Republican terms and Democrat terms)

## Cleaning

While there were outliers, those were left in the data to try and determine whether they were included in the relationship between drought severity and presidential administration. For example, they generally occurred during what appeared to be a dry season (Weeks 40-53) and at the end of a term. This is something that we would want to take into consideration when reviewing the data as there may be a significant relationship between that occurrence and the administration type. There were not a significant number of outliers, and they primarily presented themselves within the data on Republican administrations.

Beyond outliers however, when initially reviewing the data, I realized that there were regions where droughts seemed to occur regardless, one of those regions specifically being the West. From there, I decided that it would be best to look at the areas that were least likely to be affected by severe droughts and only analyze those. That brought the climate region number down from 6 to 3. The three selected climate regions were the Midwest, the Southeast, and the Northeast. When compiling my final data information, I also did not include areas where no droughts were present. I only considered areas that were affected by a D0 – D4 drought. This narrowed down my total observations to approximately 6,360.

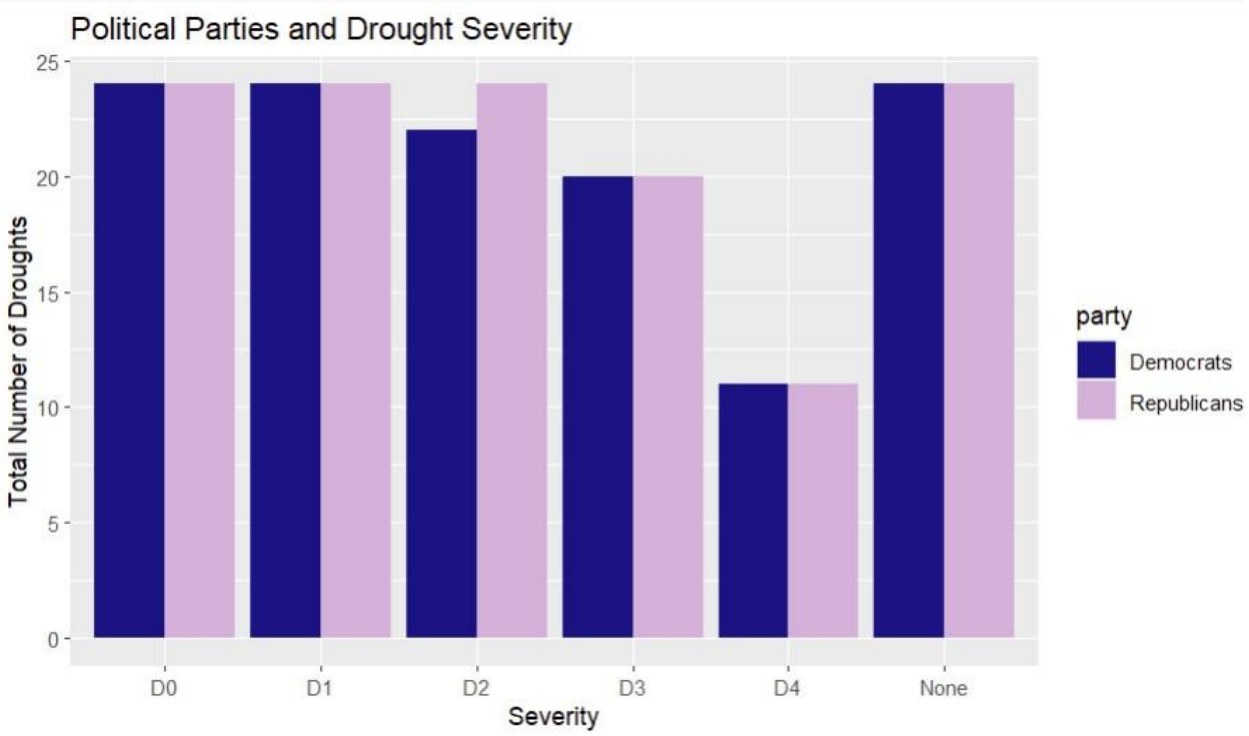
Data Analysis

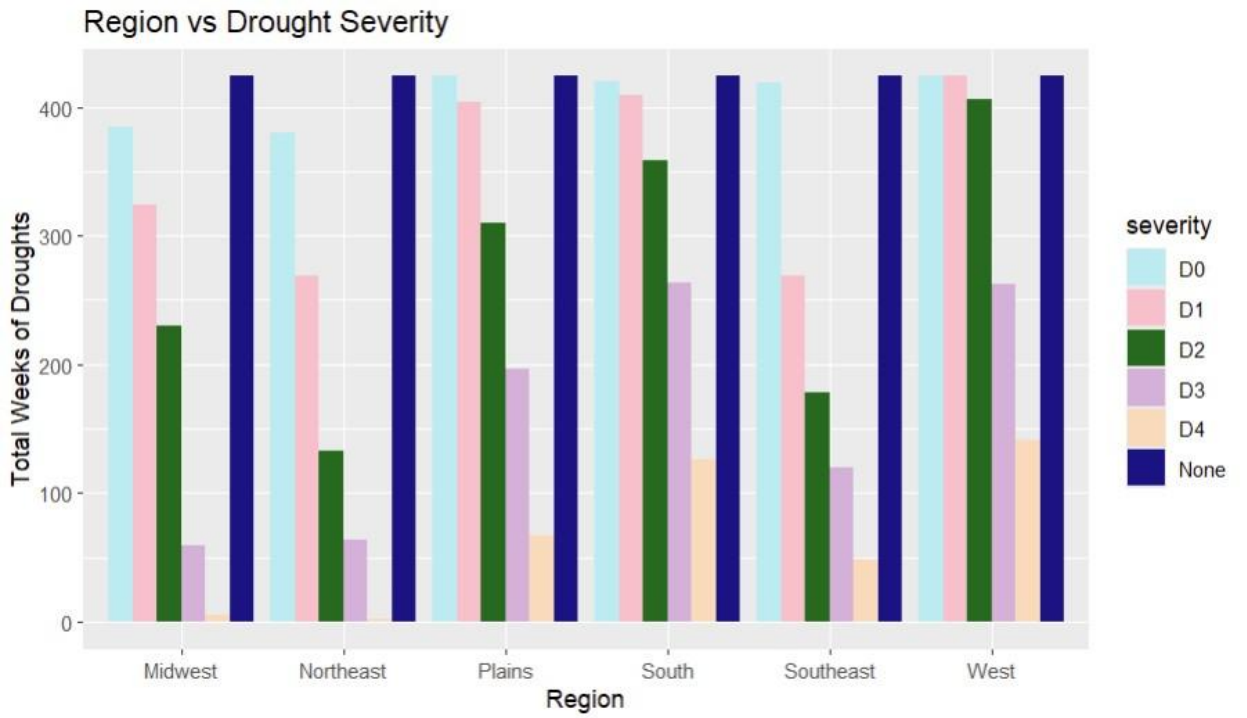
**Hypothesis**: There is a statistically significant relationship between US Executive Administration and the severity of droughts in the United States.

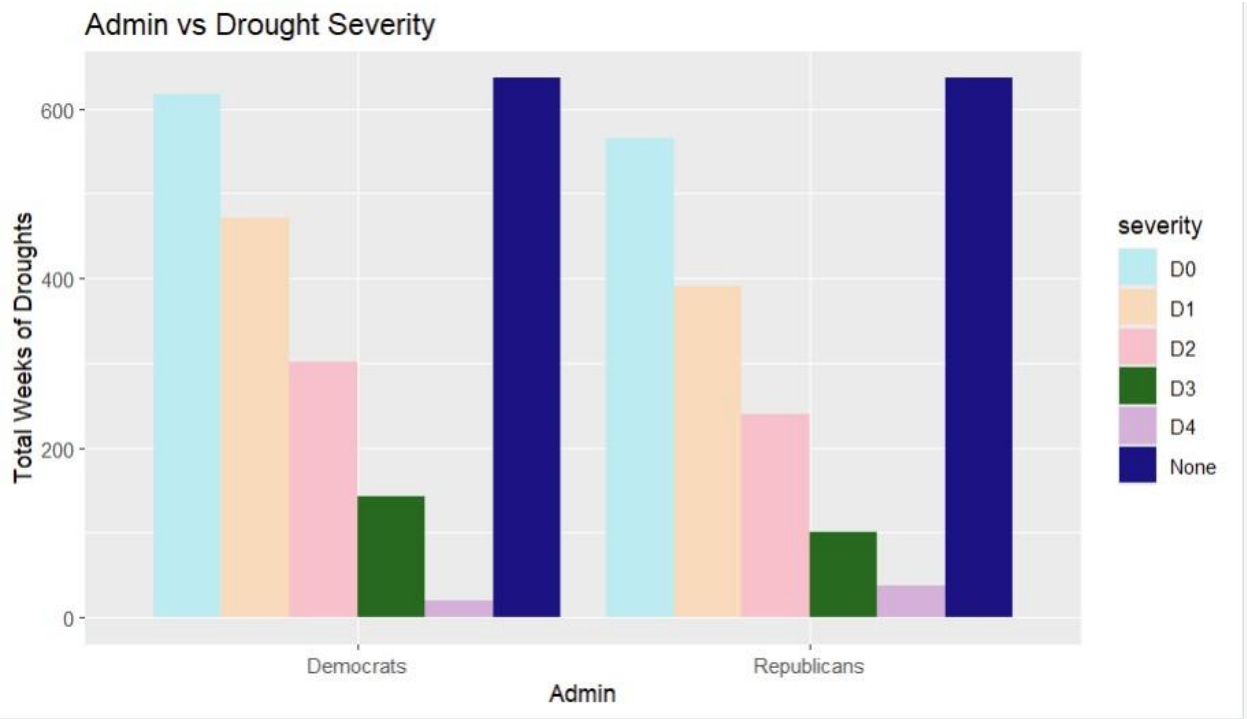
| severity | party Democrats | Republicans |
|---|---|---|
| D0 | 1250 | 1202 |
| D1 | 1078 | 1021 |
| D2 | 813 | 802 |
| D3 | 509 | 457 |
| D4 | 246 | 154 |
| None | 1272 | 1272 |

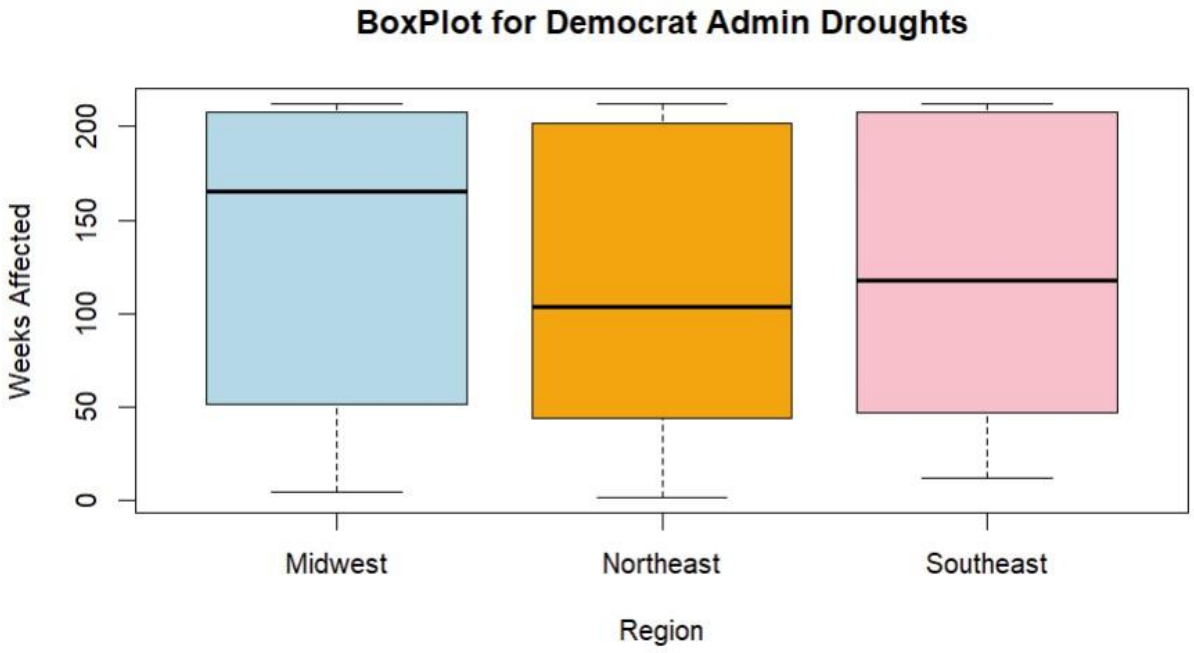
>

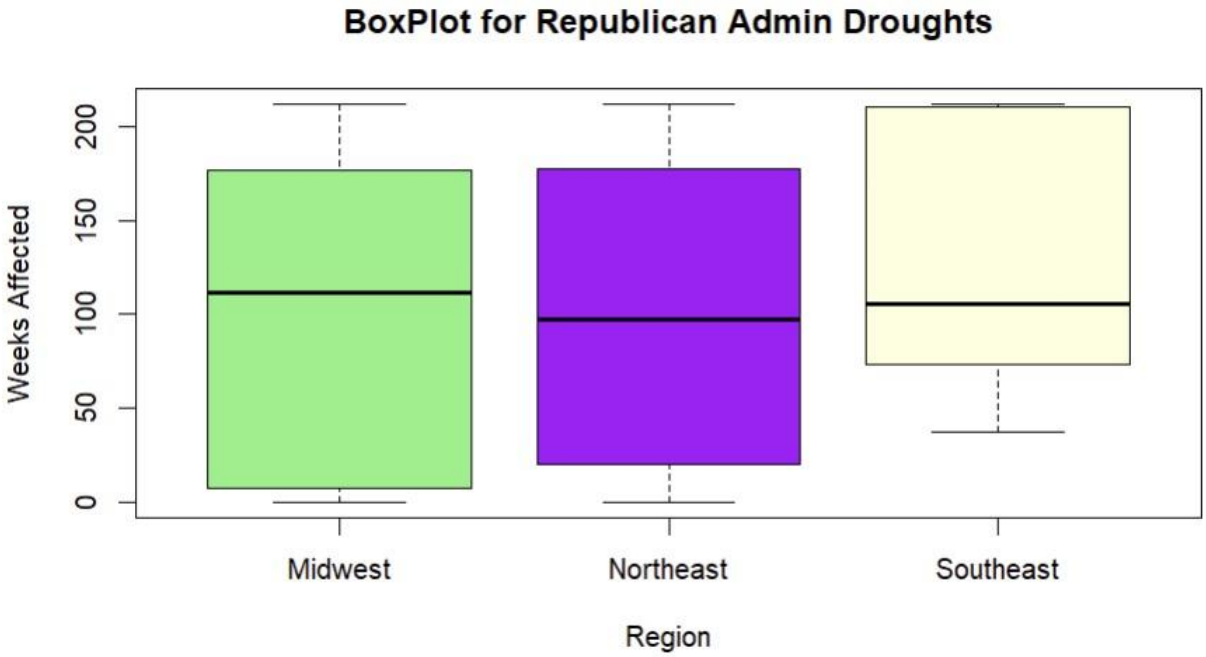
| severity | party Democrats | Republicans |
|---|---|---|
| D0 | 24 | 24 |
| D1 | 24 | 24 |
| D2 | 22 | 24 |
| D3 | 10 | 20 |
| D4 | 8 | 9 |
| None | 24 | 24 |

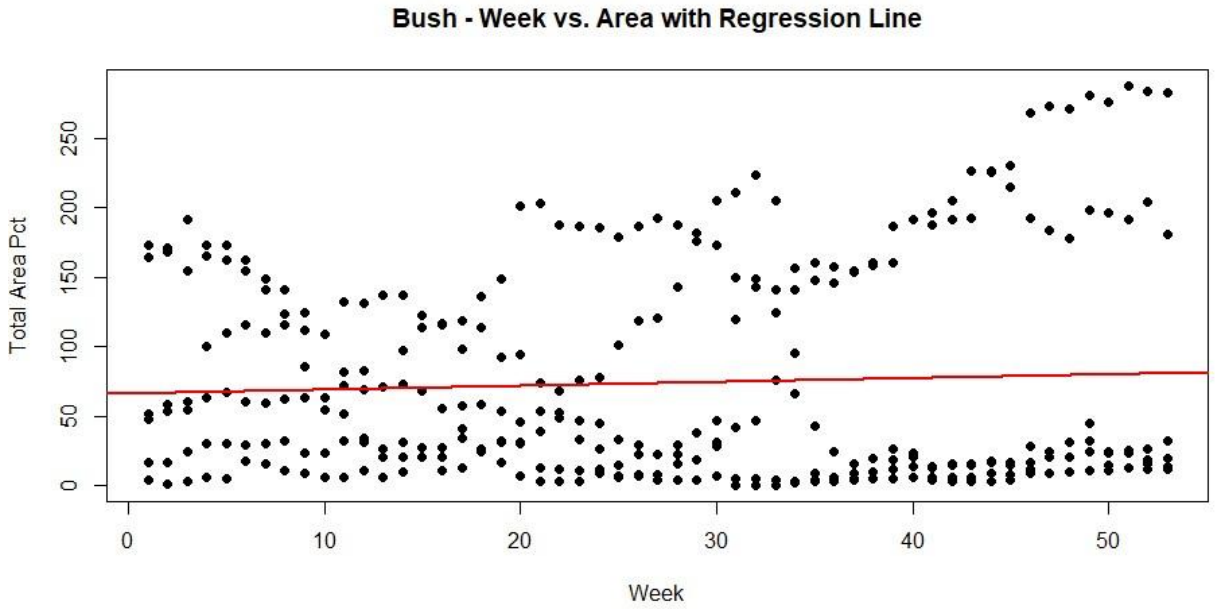


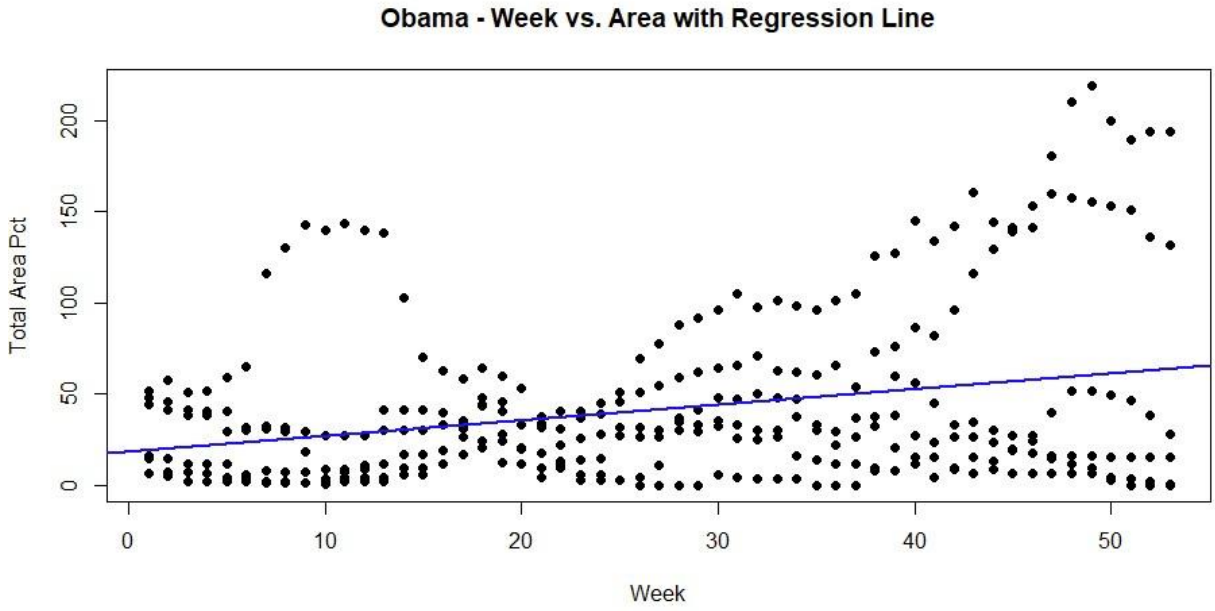

Region vs Drought Severity
Total Weeks of Droughts
0
100
200
300
400
Midwest
Northeast
Plains
South
Southeast
West
Region
severity
D0
D1
D2
D3
D4
None

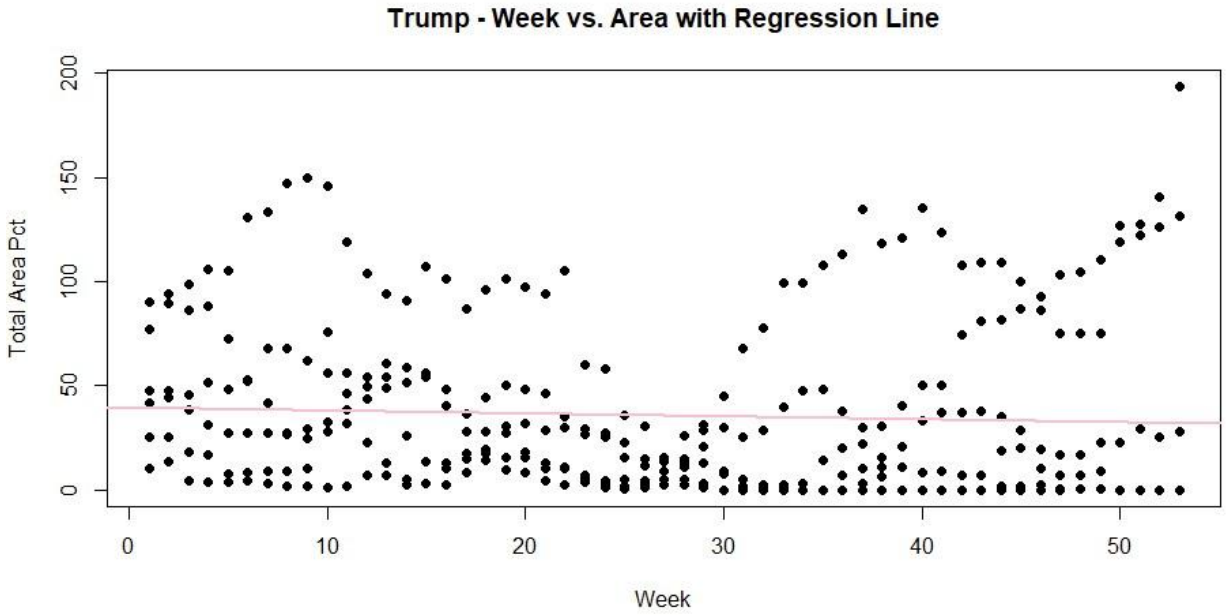

Admin vs Drought Severity
Total Weeks of Droughts
0
200
400
600
Democrats
Republicans
Admin
severity
D0
D1
D2
D3
D4
None

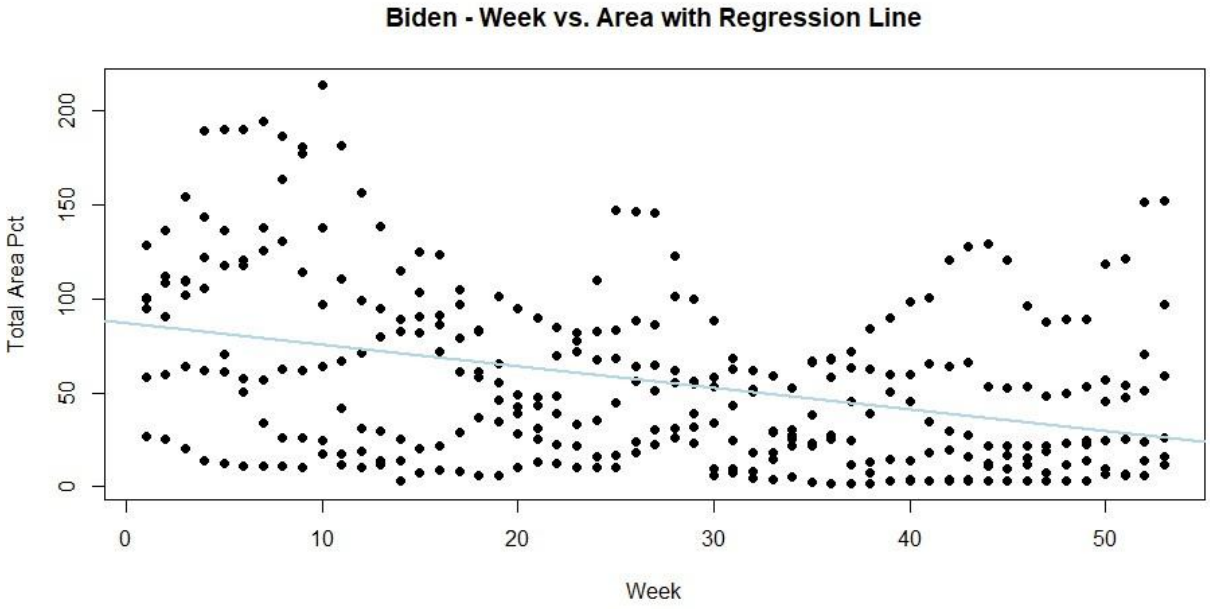

BoxPlot for Democrat Admin Droughts
Weeks Affected
0
50
100
150
200
Midwest
Northeast
Southeast
Region

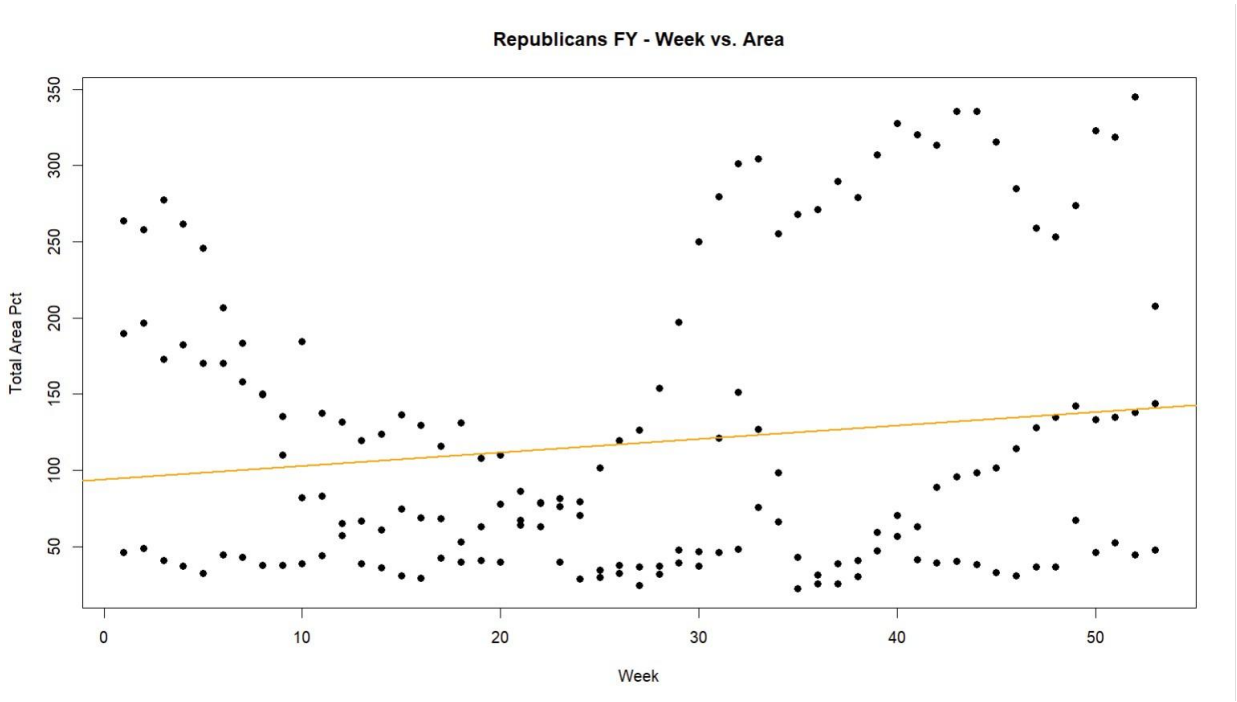

BoxPlot for Republican Admin Droughts
Weeks Affected
0
50
100
150
200
Midwest
Northeast
Southeast
Region

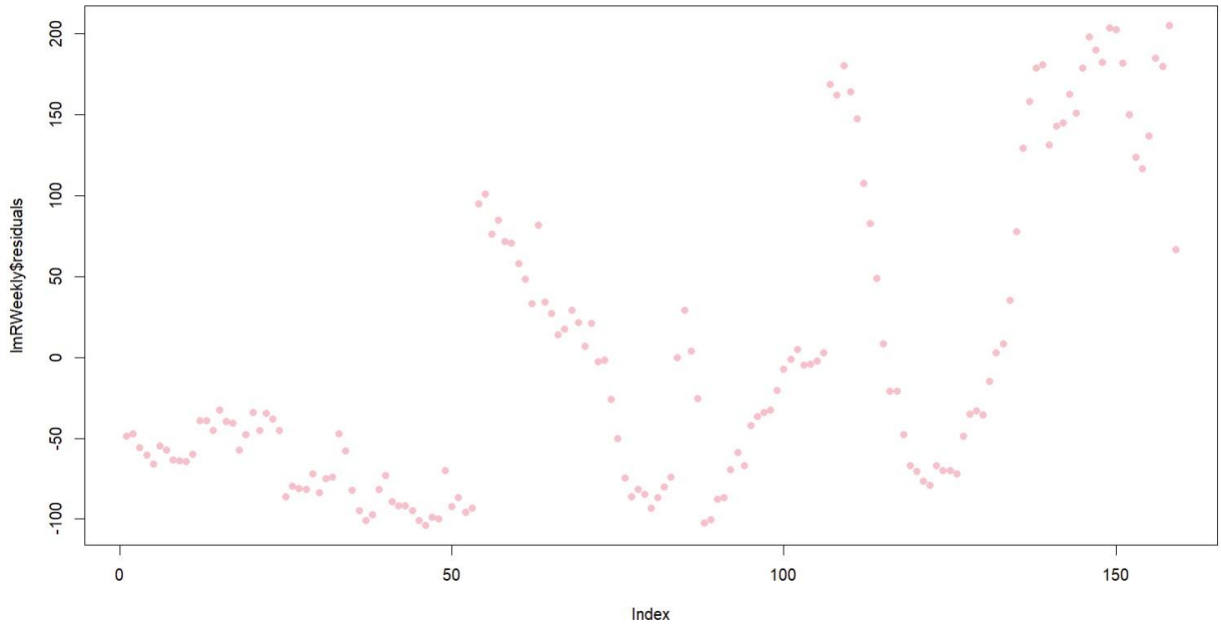

Bush - Week vs. Area with Regression Line
Total Area Pct
0
50
100
150
200
250
0
10
20
30
40
50
Week

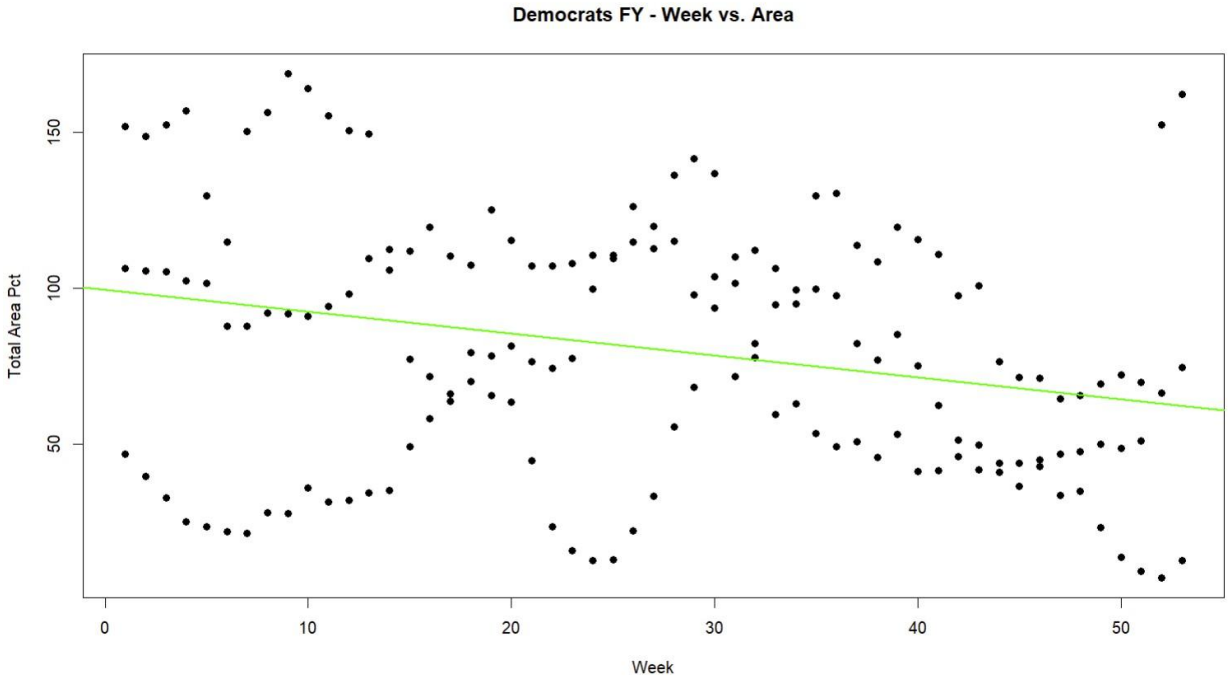

Obama - Week vs. Area with Regression Line
Total Area Pct
0
50
100
150
200
0
10
20
30
40
50
Week

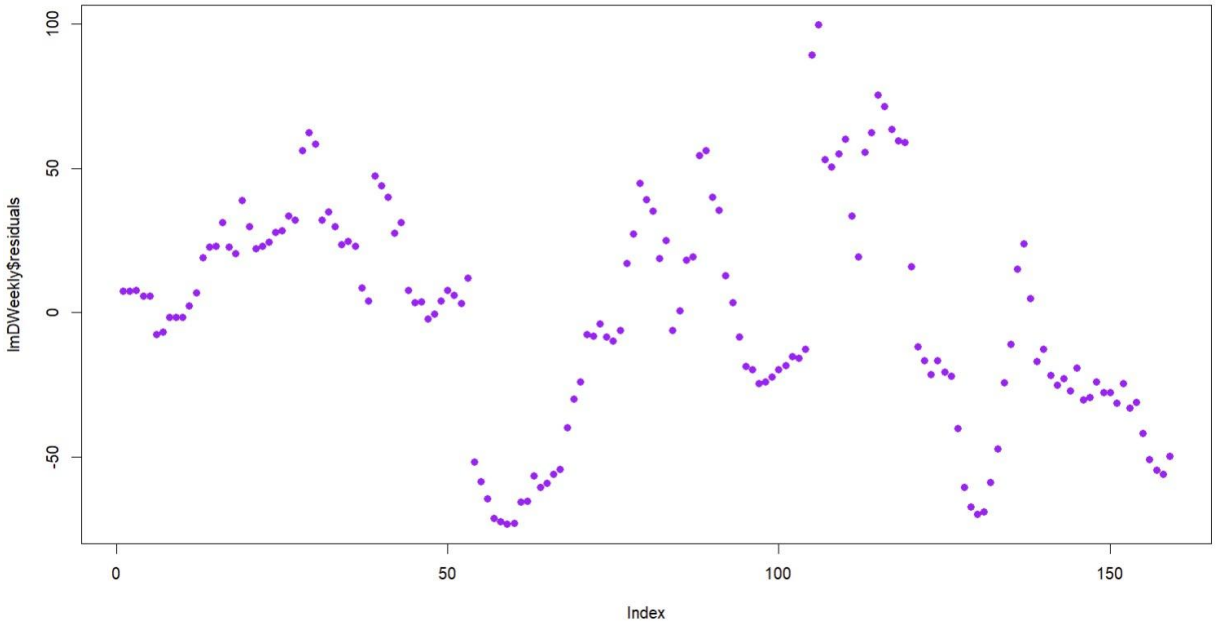

Trump - Week vs. Area with Regression Line
Total Area Pct
0
50
100
150
200
0
10
20
30
40
50
Week

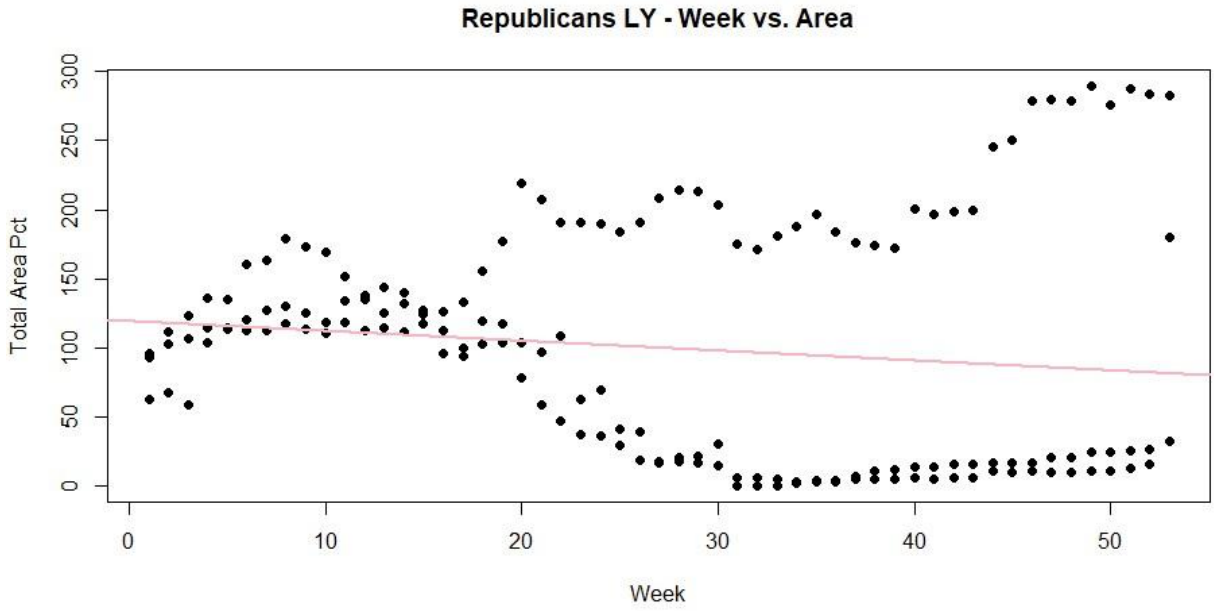

Biden - Week vs. Area with Regression Line
Total Area Pct
0
50
100
150
200
0
10
20
30
40
50
Week

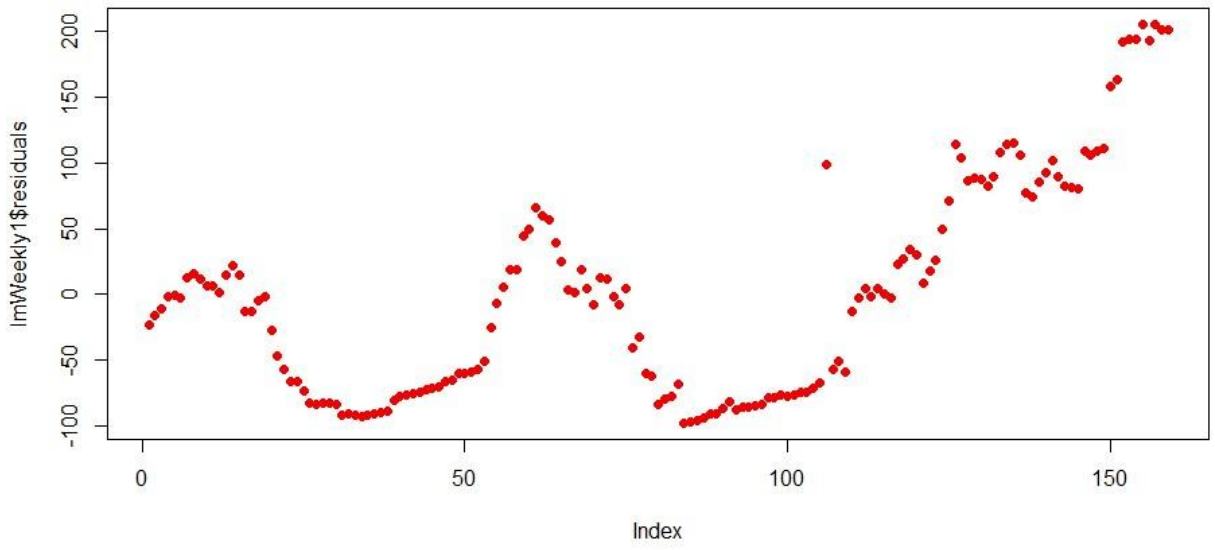

Republicans FY - Week vs. Area
Total Area Pct
Week
0
10
20
30
40
50
50
100
150
200
250
300
350

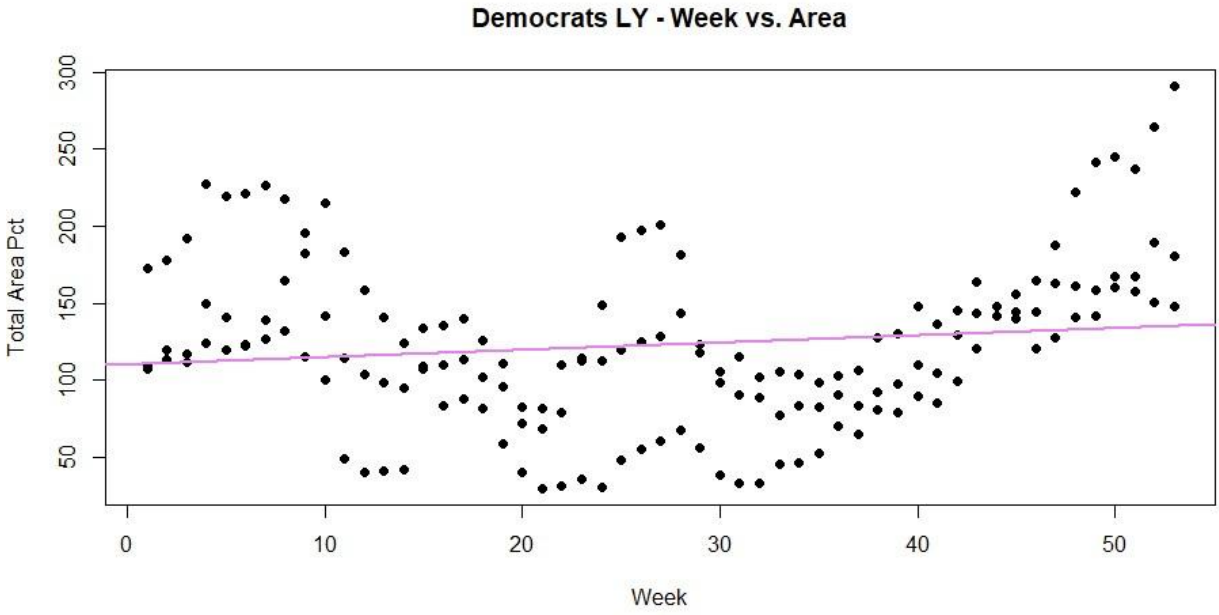

lmRWeekly$residuals
Index
0
50
100
150
-100
-50
0
50
100
150
200

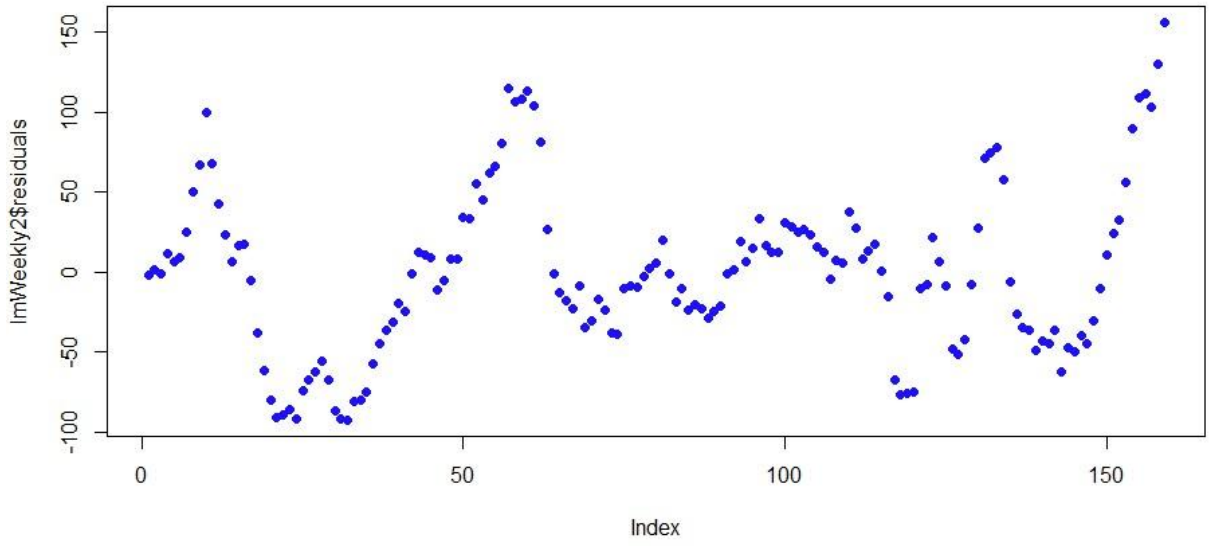

Democrats FY - Week vs. Area
Total Area Pct
Week
0
10
20
30
40
50
50
100
150

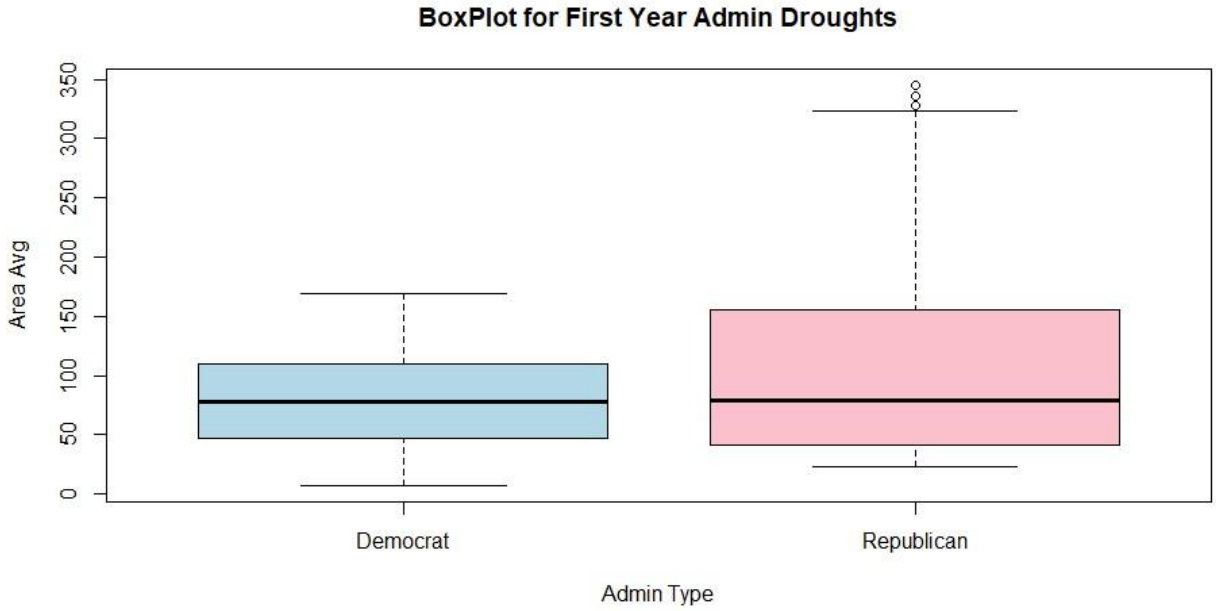

lmDWeekly$residuals
Index
0
50
100
150
-50
0
50
100

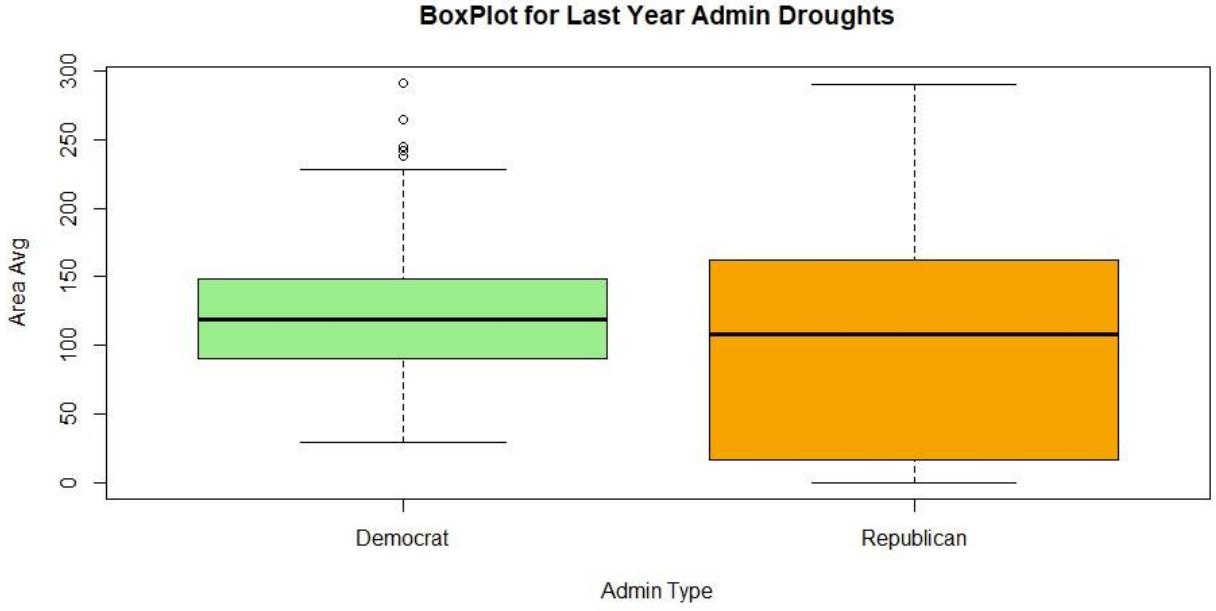

Republicans LY - Week vs. Area
Total Area Pct
0
50
100
150
200
250
300
0
10
20
30
40
50
Week


lmWeekly1$residuals
-100
-50
0
50
100
150
200
0
50
100
150
Index


Democrats LY - Week vs. Area
Total Area Pct
50
100
150
200
250
300
0
10
20
30
40
50
Week


lmWeekly2$residuals
-100
-50
0
50
100
150
0
50
100
150
Index




***Statistical Methods and EDA***

For my data, I ran four tests with a few different items. First was my chi-square test to analyze whether there was a relationship between drought severity and presidential administration. This solely focused on the variables “admin_type” and “drought severity”. This counted the total number of weeks that a certain drought severity occurred to see if a certain severity type persisted across the duration of a presidency. The null hypothesis for this test, since it was a test of independence, was that there was no relationship between the variable admin type and drought severity. The alternative hypothesis was that there was

a relationship between those two variables. After running that analysis, the p-value came out to be less than 0.05 which allowed me to conclude that there was a significant relationship between the party and the overall drought severity. This, however, did not consider what drought severity may have looked like at the beginning and end of each presidential term.

From that initial conclusion, I moved forward to break down what the data looked like in terms of a week-to-week basis for each candidate and party type. With my linear regression model, I looked at the 53-week time span for both years for the individual presidents to see if there was a relationship and then I looked solely at the first year of both candidates and the last year of both candidates. This is where we saw an even more significant relationship between the administration and drought severity by week. This also showed what the dry and wet seasons would be for the various regions being affected with peak dry conditions being at the beginning and last few weeks of the year. This is also where outliers were most likely to occur, which is why outliers remained in the data. If an outlier occurred several times, then this was something to take into consideration whereas if there was one outlier then it may genuinely be just that, an anomaly.

When reviewing the graphs, it is also important to note that while some graphs may trend upwards, such as the residuals graph for the last year comparison of Democrat and Republican administrations, the y-values are different. (I am not quite sure how to set this in R.) This led to the creation of the boxplot to put both values at the same intervals and create a better visual representation of their differences. This is where we can more clearly see how much the outliers are skewing our data if at all. For the “BoxPlot for Last Year Admin Droughts”, the outliers ran a lot further beyond the tail for the high end of our data than they did when looking at the outliers on our “BoxPlot for First Year Admin Droughts”.

***Interpretation***

Based on the data from my t tests, anova tests, linear regression models, and chi-square test, there is a significant relationship between drought severity and executive administration. There is also a significant relationship not just between administration and drought severity but also between the initial drought severity for each administration type.

The data above demonstrated that with republican administrations, we see a rise in the number of land area that is affected by droughts. We also see an influx in the number of droughts that occur of each type. With the democratic administrations however, we see that their first and last years generally trended downwards with droughts affecting less land area and becoming less frequent. When comparing the box plots between the last year of administration droughts, the outliers that we see on the Democrats data is still lower than

that of the highest tail for our Republican data. This is the primary reason that outliers were considered when plotting my data because even though outliers were present, their skew of the data still managed to show that the relationship was present and that Democratic candidates were able to lower the number of droughts comparatively.

When running our t-test on the first year and last year of Republican and Democrat data, we also found that our p-values were below 0.05, showcasing that there was still a significant relationship within those time frames.

***Limitations***

One of the biggest limitations that I faced was the fact that I was only able to analyze the first and last year of each administration. Due to this, there was not an accurate reflection overtime of the relationship being reviewed. A stronger analysis would have included a line graph that visualized the rises and falls in the percentage of land area affected by droughts on a week-by-week basis for the duration of each administrative term.

Since I limited myself to only the three regions least likely to experience droughts, the analysis did not consider whether areas that regularly struggle with droughts experienced less or less severe droughts by administration. This is something that could have also been beneficial in a full analysis of the data.

## Conclusion

I found that there is a statistically significant relationship between US Executive Administration and the severity of drought in the United States. There is a positive correlation between administration type and drought severity where severe droughts are more likely to occur during Republican administrations and less likely to occur during Democratic administrations.

For future research, I would consider the number of environmental policies that are implemented or receded during each administration and see if the droughts that occur happen around the time that those policies are implemented. I would also consider narrow down the time period of each year being evaluated to just the dry season or just the wet season (i.e. I would look at when droughts are at their peak during the winter months or when droughts are at their lowest during summer months.)